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History-Conditioned SAC under Sensor Faults

A Pilot Study of Adaptation and Retention in Continuous Control

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Motivation: Continual Control Under Sensor Faults

Why deployed reinforcement learning controllers fail when physical sensors degrade

1. Sensor Degradation

- Real robots face sudden observation noise and packet latency.
- A 3-step delay breaks the Markov assumption.
- Frozen policies suffer severe return collapse.

2. History Conditioning

- Causal Transformer encodes a 32-step observation-action window.
- Can infer velocity & delay dynamics from history.
- Higher capacity also increases overfitting risk.

3. Retention Dilemma

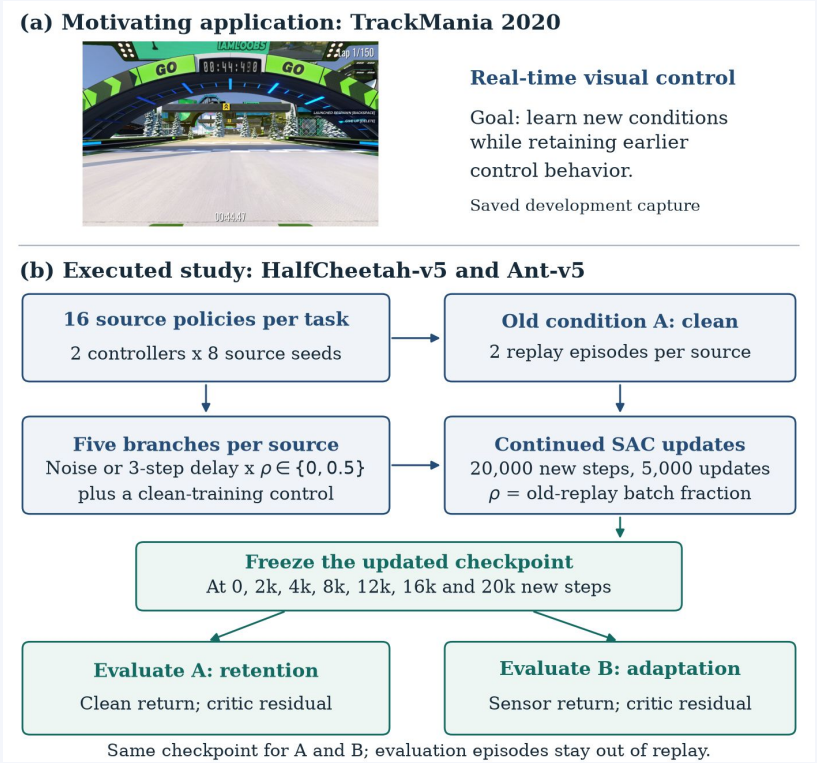
- Online fine-tuning adapts to Fault Regime B.
- However, policies overwrite clean-sensor skills (Regime A).
- Can 50% clean replay shield retention without halting adaptation?

Core Research Question: How do sequence memory (Transformer vs. Reactive MLP) and clean replay mixing ($\rho \in \{0, 0.5\}$) interact across sensor noise and observation delay?

Methodology: From Real-Time Racing to Controlled Benchmarks

Why We Decoupled Evaluation

- Motivated by high-speed telemetry faults in TrackMania 2020 autonomous driving.
- Real-time game loops introduce OS scheduling jitter that confounds paired continual RL comparisons.
- We isolate exact fault physics in MuJoCo (HalfCheetah-v5 & Ant-v5) with deterministic seed pairing.
- Every source checkpoint branches into 10 controlled adaptation arms.



Experimental Design: 160 Continual Adaptation Branches

Rigorous paired-seed factorial design across two morphologies (N = 8 seeds: 42–49 per morphology)

Policy Architectures

- Causal Transformer:
  Context length L = 32
  2 layers, 4 heads, d = 128
- Reactive MLP:
  Memoryless [256, 256]
- Shared SAC backbone:
  Twin Q-critics, auto α

Fault Regimes (B)

- Clean Source (A):
  300k steps pretraining
- Sensor Noise (B1):
  Gaussian σ = 0.10
- Sensor Delay (B2):
  k = 3 steps lag
- Clean Continuation:
  Drift control on A

Replay & Metrics

- 20,000 steps / branch
  (5,000 gradient updates)
- ρ = 0.0 (Pure Fault B)
- ρ = 0.5 (50% Clean A)
- Metrics: ΔB (Adaptation),
  ΔA (Clean Retention),
  Wilcoxon + Holm-Bonferroni

Total Scale: 16 Pretrained Source Checkpoints × 10 Branches = 160 Continual Adaptation Runs (HalfCheetah-v5 & Ant-v5)

Core Finding #1: Replay Shields Against Catastrophic Forgetting

Under 4-step sensor delay on HalfCheetah-v5, pure online adaptation (ρ = 0) destroys clean-sensor return

-2,496

Transformer Clean Retention Loss (ΔA at ρ = 0)
Drops from 4,347 → 1,852 without clean replay
(Reactive MLP also drops ΔA = -1,390)

+1,323

Transformer Paired Replay Rescue (D_ΔA at ρ = 0.5)
95% CI [595, 2,050] • 8/8 Seeds Positive!
Reactive Paired Rescue: +1,406 (8/8 Seeds Positive)

Control Verification: Clean-continuation controls improve (+782 Transformer, +963 Reactive), proving that forgetting is strictly caused by delayed-sensor adaptation rather than training instability.

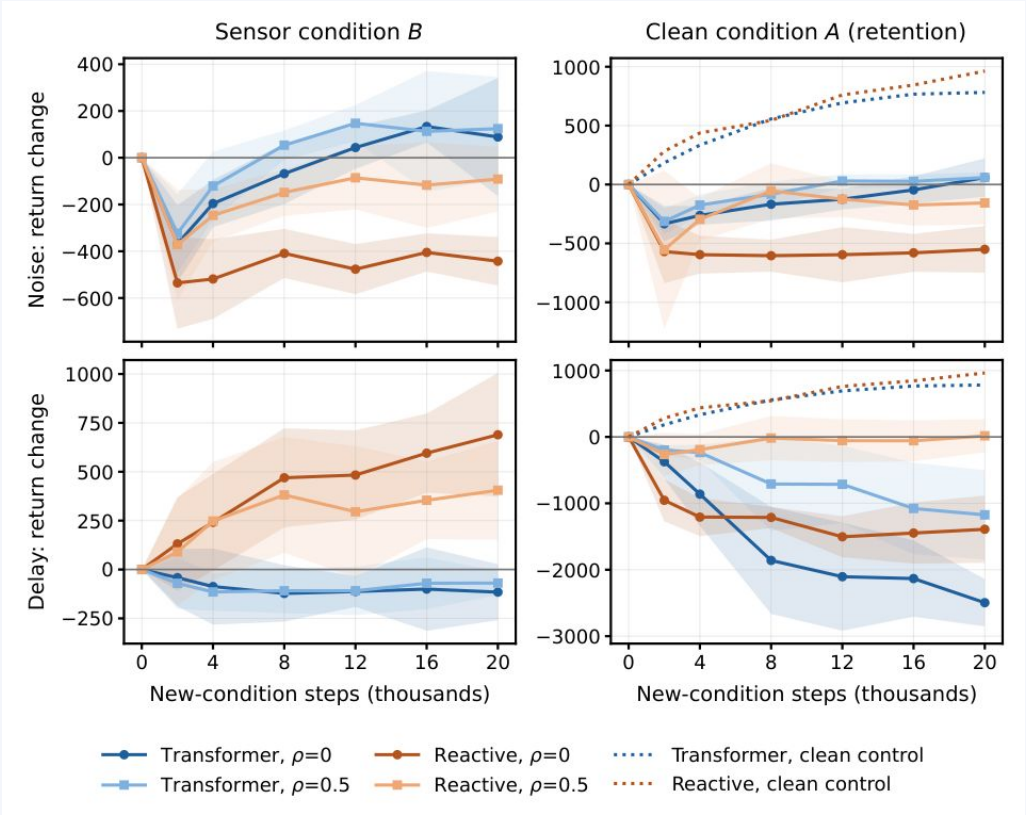
HalfCheetah-v5: Adaptation & Retention Curves

Key Observations (N = 8 Seeds)

• Delay Adaptation (ΔB):
Both architectures recover +1,630 to +2,045 return under 3-step lag.

• Retention Collapse (ρ=0):
Dashed clean-evaluation curves plunge sharply during delay adaptation.

• Replay Synergy in Noise:
For Reactive under noise, ρ=0.5 boosts both fault adaptation (+351, 8/8) and retention (+394, 8/8).



HalfCheetah-v5: Paired Replay Effect Summary (N = 8)

Architecture & Fault	ΔB (ρ=0)	ΔB (ρ=0.5)	D_ΔB (+Seeds)	ΔA (ρ=0 → 0.5)	D_ΔA (+Seeds)
Transformer • Delay	+1,630	+1,813	+183 (6/8)	-2,496 → -1,173	**+1,323* (8/8)**
Reactive MLP • Delay	+2,045	+1,761	-284 (1/8)	-1,390 → +16	**+1,406* (8/8)**
Transformer • Noise	+588	+461	-126 (4/8)	+130 → +470	+340 (6/8)
Reactive MLP • Noise	+242	+593	**+351* (8/8)**	+237 → +631	**+394* (8/8)**

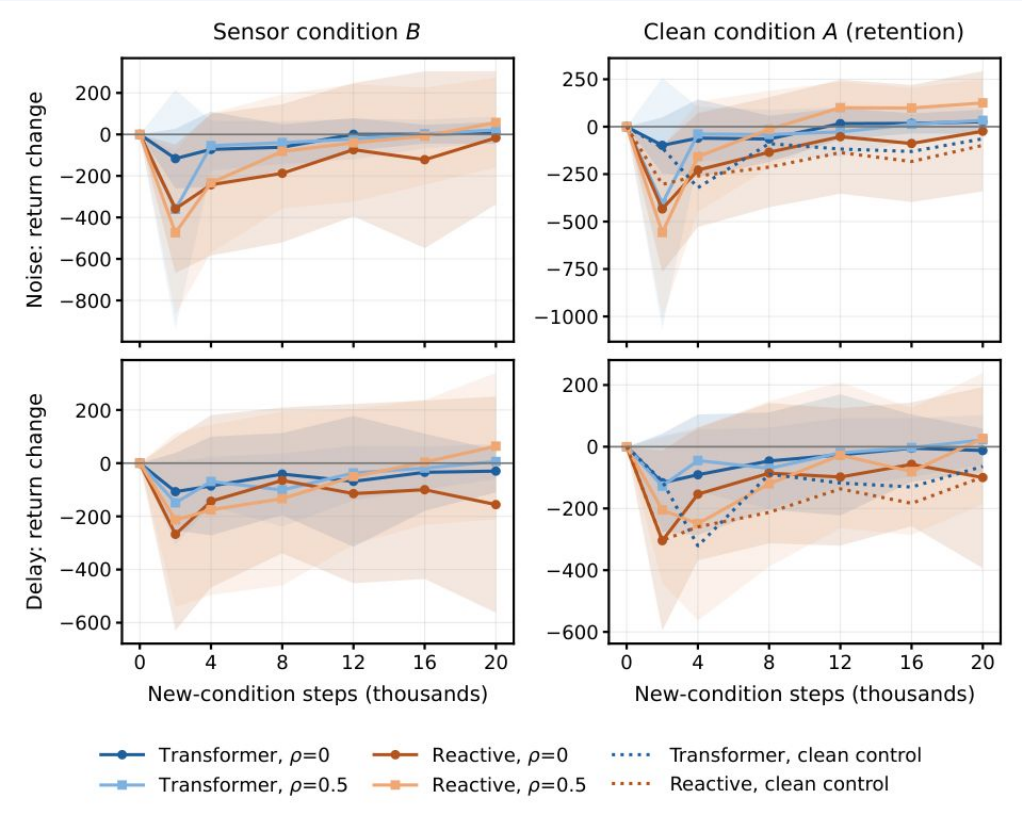
*** All 8/8 paired seeds positive: exact Wilcoxon signed-rank p_raw = 0.0078 (Holm-Bonferroni adjusted p_Holm = 0.0625 across 8 tests; paired t-test p < 0.004).**

Cross-Morphology Generalization: Ant-v5 (27-D State)

3D Quadruped Findings (80 Runs)

• Reactive Delay Synergy:
Adding 50% clean replay boosts Ant-v5 delay adaptation from +308 → +915 (D_ΔB = +607, 7/8 seeds) and rescues clean retention from -320 → +99 (D_ΔA = +418, 7/8 seeds).

• High-Dimensional Sequence Bottleneck:
On 27-D Ant-v5, 32-step Transformer requires longer pretraining horizon; clean-replay anchors value targets during contact-rich locomotion.



Mechanistic Diagnostics: Critic Bias & 500 Hz Latency

What Replay Creates a Plasticity Trade-Off — and Why Transformer Remains Real-Time Capable

1. Off-Policy Critic Distortion (Delay)

- Mixing clean (A) and 3-step delayed (B) transitions pairs one critic with two distinct transition kernels.
- Held-out Bellman Residual under Delay:
  - Transformer: 12.1 (ρ=0) → 13.7 (ρ=0.5)
  - Reactive MLP: 12.0 (ρ=0) → 14.3 (ρ=0.5)
- Value-Bias Gap (Q_pred - G_MC):
  - Stays elevated at +62.0 (Transformer) and +90.4 (Reactive) with ρ=0.5.
- Explains why Reactive delay adaptation drops slightly (-284) when forced to retain clean A.

2. Real-Time CPU Inference Latency

- Benchmarked 10,000 single-step forward passes (batch size = 1, 1 CPU thread, Ryzen 9 8940HX).
- Causal Transformer (L = 32):
  - Mean: 0.58 ms | Median: 0.56 ms
  - p99: 0.66 ms | Max: 1.56 ms
- Reactive MLP:
  - Mean: 0.09 ms | p99: 0.13 ms
- Both easily satisfy the 2.0 ms real-time budget required for 500 Hz robotic & telemetry loops!

BINUS e-PDP Reporting Package (Paper #193 Checklist)

All 7 required reimbursement items staged in output/binus_epdp_reporting_paper193/ for SharePoint upload

# Required Document	Prepared File / Artifact in Workspace	Status / Action
1. Scopus Indexed Evidence	1_Scopus_Indexed_Evidence/icodit_2026_scopus_evidence.png & .pdf	**Ready (Web capture)**
2. Conference Invoice	2_Invoice_From_Conference/invoice_icodit_2026_paper193.pdf	**Downloaded from Drive**
3. Draft Paper (Pre-Revision)	3_Draft_Paper_Before_Revision/icodit-2026-anonymous-review-draft.pdf	**Ready (Anonymous draft)**
4. Open Data Package	4_Open_Data/paper193_open_data_160_runs.zip (CSVs, JSONs, Curves)	**Ready (Verified)**
5. Open Contribution (Code)	5_Open_Contribution/paper193_open_code_and_protocol.zip	**Ready (Scripts & Docs)**
6. Final Version Paper	6_Final_Version_Paper/ICoDiT2026_Paper193_CameraReady.pdf (15 pp.)	**Ready (LNCS Format)**
7. Letter of Acceptance (LoA)	7_Letter_Of_Acceptance/ICoDiT2026_Paper193_LoA.pdf	**Compiled from Official Emails**

Conclusions & Takeaways

1. 50% Clean Replay Prevents Catastrophic Forgetting: Rescues clean-regime return in 8/8 HalfCheetah seeds (+1,323 Transformer, +1,406 Reactive) and 7/8 Ant seeds (+418) under 3-step sensor delay.

2. Critic Distortion Explains the Plasticity–Stability Trade-Off: Mixing clean and delayed transitions inflates soft Bellman residuals, motivating fault-context-conditioned critics.

3. Real-Time Ready at 500 Hz: Causal Transformer (L=32) executes in 0.58 ms mean (0.66 ms p99) on a single CPU thread.

Thank You! • Questions & Discussion

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